01 / PROBLEM AND CONTRIBUTION

8051 gas-leak detection & alerting

Microcontrollers mini-project | Undergraduate | Team project

RESULT AT A GLANCE

Documented normal and alarm-state behavior for an 8051-based controller, connecting sensor input to display, LED/buzzer alerts, and fan activation.

The engineering question

How can a microcontroller translate a gas-detection input into visible and audible alerts?

Situation

A gas-detection input must be translated into a clear alarm state and output controls. The academic project explores that path with an AT89S52, controller code, and circuit simulation.

Task

Contribute to the team's microcontroller implementation and documented normal/alarm-state behavior using the supplied 8051 and Proteus materials.

My contribution

Completed as a team microcontroller project, with controller code, circuit simulation, and a shared technical report.

Tools and concepts

8051 / Assembly / Keil µVision / Proteus / Sensor interfacing

02 / DESIGN AND METHOD

From the task to implementation

01 / Implementation step

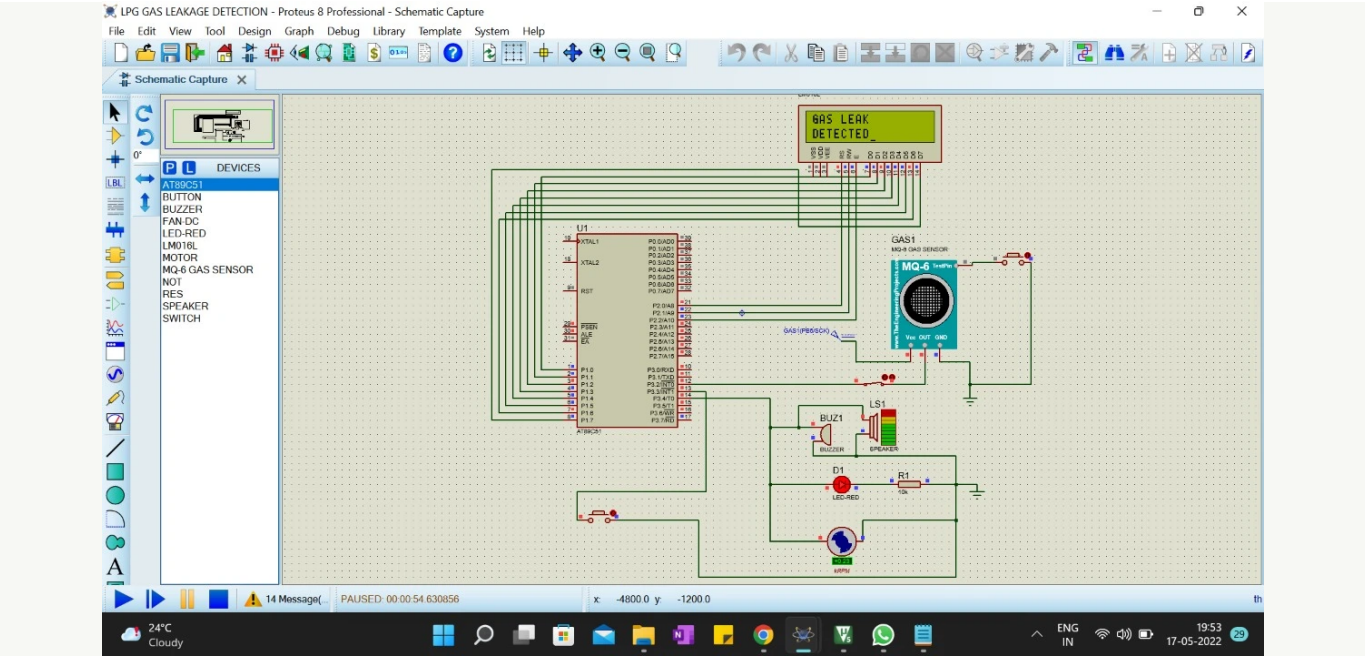
Developed the controller flow around the sensor input, LED/buzzer outputs, and fan activation. The report identifies the AT89S52 variant and includes 8051 assembly.

02 / Implementation step

Used Keil tooling for the controller program; the supplied project folder also contains the Proteus circuit project and simulation material.

Controller behavior

The AT89S52 program connects the sensor input condition to normal/alarm messages and controlled outputs. Keil and Proteus materials support the controller and circuit workflow; the project remains the documented 8051 version.



8051 gas-alarm simulation

Proteus schematic from the AT89S52 gas-detection report.

03 / RESULTS AND ENGINEERING PERSPECTIVE

Evidence, interpretation, and next steps

Reported result

The report and source files document normal and alarm-state behavior, display messages, and output control.

Beyond logical alarm response

A correct branch to the alarm state does not establish a calibrated gas-concentration threshold. Environmental sensitivity, sensor calibration, and practical-device testing require separate characterization.

Scope and limitations

The project explores the sensor-to-alert control path in an academic prototype. Sensor calibration and environmental sensitivity would need further characterization for a practical device.

Possible next steps - proposed, not completed results

- Document the input threshold, sensor calibration procedure, and test environment.
- Add controlled normal/alarm transition tests and recovery behavior without claiming safety certification.

Source material: LPG GAS LEAKAGE DETECTION USING 8051 MICROCONTROLLERS.docx and associated Keil/Proteus project files.

Prepared from the supplied reports, project files, and verified portfolio copy. Figures retain their original content; explanatory diagrams are labelled.

Project files and documentation: LPG-Gas-leakage-detector-using-8051-Microcontroller